

09 Model Training and Evaluation

Course Name: Data Management for Machine Learning

Assignment Title: End-to-End Data Management Pipeline for a Recommendation System

Assignment: Group 51 - Data management for Machine Learning Group 51

Team Members

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1. Objective

This report captures the fast model training and evaluation workflow for the recommendation system, including model training, offline ranking metrics, and tracked run metadata.

2. Performance Optimizations Applied

- Only targeted folders are scanned instead of the full repository.
- Only required columns are loaded from datasets.
- Interaction rows are capped at 250,000.
- Product rows are capped at 30,000.
- Evaluation users are capped at 1,500.
- PDF content is limited to small previews and summary tables only.

3. Artifacts

Artifact	Relative Path	Last Modified	Size
run_validation.txt	src\02-validation\run_validation.txt	2026-04-28 20:14:12	46.94 KB
data_quality_report_20260429T113422Z.json	reports\validation\data_quality_report_20260429T113422Z.json	2026-04-29 16:47:43	6.66 KB
validation_log_20260429T113422Z.jsonl	logs\validation_log_20260429T113422Z.jsonl	2026-04-29 16:47:44	0.66 KB
model_training_log_20260430T124222Z.jsonl	logs\model_training_log_20260430T124222Z.jsonl	2026-04-30 18:12:57	0.38 KB
model_run_20260430T124222Z.json	models\tracking\model_run_20260430T124222Z.json	2026-04-30 18:12:57	1.49 KB
model_runs_registry.csv	models\tracking\model_runs_registry.csv	2026-04-30 18:12:57	0.41 KB

Artifact	Relative Path	Last Modified	Size
model_metrics_20260430T124222Z.csv	models\reports\model_metrics_20260430T124222Z.csv	2026-04-30 18:12:57	0.22 KB
per_user_metrics_20260430T124222Z.csv	models\reports\per_user_metrics_20260430T124222Z.csv	2026-04-30 18:12:57	378.58 KB
sample_recommendations_20260430T124222Z.csv	models\reports\sample_recommendations_20260430T124222Z.csv	2026-04-30 18:12:57	1.74 KB
collaborative_svd_20260430T124222Z.pkl	models\artifacts\collaborative_svd_20260430T124222Z.pkl	2026-04-30 18:12:32	8356.2 KB
content_tfidf_20260430T124222Z.pkl	models\artifacts\content_tfidf_20260430T124222Z.pkl	2026-04-30 18:12:35	2081.07 KB

4. Training Assets

Training Asset	Relative Path	Last Modified	Size
No src training asset found	-	-	-

5. Data Summary

dataset	rows	columns
events_source	196263	5
train_split	61663	5
test_split	24681	5
products_source	194	6

6. Run Summary

run_id	tracking_tool	collaborative_status	content_status	eval_user_cap	event_row_cap
20260430T124222Z	custom_local_registry_equivalent_to_mlflow	TRAINED	TRAINED	1500	250000

7. Training Notes

- Standardized interaction columns.
- Prepared 196263 interaction rows.
- Prepared 194 product rows.
- Train rows: 61663 | Test rows: 24681
- Trained collaborative filtering using TruncatedSVD.
- Trained content-based filtering using TF-IDF.

8. Model Performance Report

model_name	users_evaluated	precision_at_k	recall_at_k	map_at_k	ndcg_at_k	hit_rate_at_k
collaborative_svd	1500	0.001733	0.017333	0.008948	0.010922	0.017333
content_tfidf	1500	0.000933	0.009333	0.003789	0.005117	0.009333

9. Per-User Evaluation Preview

model_name	user_id	true_item	recommended_items	hit	rank	precision_at_k	recall_at_k	map_at_k	ndcg_at_k
collaborative_svd	1001673	187026	122219, 319942, 378076, 329279, 333089, 1891, 314789, 387747, 288418, 448494	0		0.0	0.0	0.0	0.0
collaborative_svd	1001857	254418	89044, 205139, 463280, 463027, 303715, 334401, 133647, 387373, 181527, 315543	0		0.0	0.0	0.0	0.0
collaborative_svd	100283	392194	421884, 441415, 163689, 370745, 118199, 387318, 366888, 418795, 126019, 189444	0		0.0	0.0	0.0	0.0
collaborative_svd	100398	141625	441415, 60980, 463280, 255708, 100393, 370745, 175485, 71733, 187359, 179601	0		0.0	0.0	0.0	0.0
collaborative_svd	100475	108096	421884, 464142, 362697, 179601, 463027, 199348, 441415, 38706, 463280, 15822	0		0.0	0.0	0.0	0.0
collaborative_svd	1005041	149571	175356, 23762, 241555, 177522, 76196, 158666, 207546, 352258, 279457, 354676	0		0.0	0.0	0.0	0.0
collaborative_svd	1005983	287871	355994, 463663, 331311, 336121, 457045, 122604, 422671, 306763, 66451, 340960	0		0.0	0.0	0.0	0.0

model_name	user_id	true_item	recommended_items	hit	rank	precision_at_k	recall_at_k	map_at_k	ndcg_at_k
collaborative_svd	100643	218871	420960, 179601, 170262, 286731, 119736, 222604, 8523, 268883, 301721, 262477	0		0.0	0.0	0.0	0.0
collaborative_svd	1006760	200799	243980, 257040, 318610, 309778, 420960, 397692, 436115, 212216, 442131, 51792	0		0.0	0.0	0.0	0.0
collaborative_svd	1007190	158507	355994, 286731, 324109, 50648, 113233, 268883, 130371, 260256, 242521, 218626	0		0.0	0.0	0.0	0.0
collaborative_svd	100735	249649	460553, 5411, 27537, 276704, 360487, 243980, 428648, 397424, 249207, 165030	0		0.0	0.0	0.0	0.0
collaborative_svd	1008246	172894	354881, 247240, 411886, 257040, 366177, 119736, 425920, 57788, 151178, 75820	0		0.0	0.0	0.0	0.0

10. Sample Recommendations

user_id	heldout_item	collaborative_top_k	content_top_k
10002	453618	5411, 7943, 309778, 372188, 8523, 49692, 380014, 420960, 380810, 367664	309778, 5411, 257040, 234255, 268335, 303715, 360487, 194791, 461686, 367664
1000264	18447	355994, 369158, 122604, 463663, 382958, 175485, 441415, 134255, 187359, 60323	309778, 5411, 257040, 234255, 268335, 303715, 360487, 194791, 461686, 367664
1000294	190181	5411, 175356, 400077, 442725, 456056, 329133, 158666, 279457, 367664, 354676	309778, 5411, 257040, 234255, 268335, 303715, 360487, 194791, 461686, 367664
1000346	181526	257040, 247240, 423339, 354881, 411886, 306034, 315685, 175485, 442131, 11311	309778, 5411, 257040, 234255, 268335, 303715, 360487, 194791, 461686, 367664
1000562	289976	460553, 369158, 325310, 241555, 184591, 400077, 148446, 324670, 86057, 96423	309778, 5411, 257040, 234255, 268335, 303715, 360487, 194791, 461686, 367664
100058	389615	463663, 355994, 179601, 122604, 60323, 331311, 336121, 134255, 457045, 66451	309778, 5411, 257040, 234255, 268335, 303715, 360487, 194791, 461686, 367664
1000640	357579	243980, 430550, 257040, 318610, 397692, 212216, 5411, 89333, 411886, 436115	309778, 5411, 257040, 234255, 268335, 303715, 360487, 194791, 461686, 367664
1000688	196345	309778, 257040, 243980, 420960, 399031, 140861, 29196, 430550, 268883, 318610	309778, 5411, 257040, 234255, 268335, 303715, 360487, 194791, 461686, 367664
1000709	76731	257040, 391512, 463663, 27537, 355994, 77769, 165030, 276704, 397424, 235636	309778, 5411, 257040, 234255, 268335, 303715, 360487, 194791, 461686, 367664
1000844	50357	309778, 257040, 140861, 277119, 150882, 318333, 249207, 7943, 288525, 2836	309778, 5411, 257040, 234255, 268335, 303715, 360487, 194791, 461686, 367664

11. Tracked Run Metadata Preview

```
{
  "run_id": "20260430T124222Z",
  "run_ts": "2026-04-30T12:42:22.629124+00:00",
  "tracking_tool": "custom_local_registry_equivalent_to_mlflow",
  "events_source": "data\\prepared\\prepared_interactions_20260429_215532.parquet",
  "products_source": "data\\bronze\\dummyjson\\products.parquet",
  "source_notes": [
    "Standardized interaction columns.",
    "Prepared 196263 interaction rows.",
    "Prepared 194 product rows."
  ],
  "split_notes": [
    "Train rows: 61663 | Test rows: 24681"
  ],
  "collaborative_status": "TRAINED",
  "content_status": "TRAINED",
  "collaborative_params": {
    "n_components": 16,
    "n_users": 24681,
    "n_items": 28837,
    "top_k": 10
  },
  "content_params": {
    "max_features": 3000,
    "ngram_range": "1,1",
    "n_items_indexed": 13,
    "top_k": 10
  },
  "collaborative_artifact": "models\\artifacts\\collaborative_svd_20260430T124222Z.pkl",
  "content_artifact": "models\\artifacts\\content_tfidf_20260430T124222Z.pkl",
  "metrics": [
    {
      "model_name": "collaborative_svd",
      "users_evaluated": 1500,
      "precision_at_k": 0.001733,
      "recall_at_k": 0.017333,
      "map_at_k": 0.008948,
      "ndcg_at_k": 0.010922,
      "hit_rate_at_k": 0.017333
    },
    {
      "model_name": "content_tfidf",
      "users_evaluated": 1500,
      "precision_at_k": 0.000933,
      "recall_at_k": 0.009333,
      "map_at_k": 0.003789,
      "ndcg_at_k": 0.005117,
      "hit_rate_at_k": 0.009333
    }
  ]
}
```

12. Validation Preview

```
{
  "cells": [
    {
      "cell_type": "code",
      "execution_count": 2,
      "id": "fef32579-f890-4b53-8d24-2fcae64a4914",
      "metadata": {},
      "outputs": [
        {
          "name": "stdout",
          "output_type": "stream",
          "text": [
            "PROJECT_ROOT: C:\\Users\\barath\\recomart-pipeline\\n",
            "RAW_ROOT: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\n",
            "BRONZE_ROOT: C:\\Users\\barath\\recomart-pipeline\\data\\bronze\\n",
            "\\n",
            "Discovered files:\\n",
            "- events_csv: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrocket\\load_date=2026-04-28\\load_hour=14\\events.csv\\n",
            "- category_tree_csv: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrocket\\load_date=2026-04-26\\load_hour=19\\category_tree.csv\\n",
            "- item_properties_part1_csv: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrocket\\load_date=2026-04-28\\load_hour=14\\item_properties_part1.csv\\n",
            "- item_properties_part2_csv: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrocket\\load_date=2026-04-28\\load_hour=14\\item_properties_part2.csv\\n",
            "- products_raw_json: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\dummyjson\\load_date=2026-04-28\\load_hour=14\\products_raw.json\\n",
            "- categories_raw_json: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\dummyjson\\load_date=2026-04-28\\load_hour=14\\categories_raw.json\\n",
            "- products_parquet: C:\\Users\\barath\\recomart-
```

```

pipeline\\data\\bronze\\dummyjson\\products.parquet\\n",
"- categories_parquet: C:\\Users\\barath\\recomart-
pipeline\\data\\bronze\\dummyjson\\categories.parquet\\n",
"\\n",
"Initial status: FAIL\\n",
"Final status: PASS\\n",
"Initial issues file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_issues_initial_20260428T144031Z.csv\\n",
"Initial summary file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_summary_initial_20260428T144031Z.csv\\n",
"Revalidation issues file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_issues_revalidated_20260428T144031Z.csv\\n",
"Revalidation summary file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_summary_revalidated_20260428T144031Z.csv\\n",
"Fix log file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\fix_log_20260428T144031Z.csv\\n",
"JSON report file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\data_quality_report_20260428T144031Z.json\\n",
"PDF report file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\data_quality_report_20260428T144031Z.pdf\\n"

```

13. Data Quality JSON Preview

```

{
  "run_id": "20260429T111342Z",
  "run_ts": "2026-04-29T11:13:42.777701+00:00",
  "project_root": "C:\\Users\\barath\\recomart-pipeline",
  "initial_validation": {
    "run_id": "20260429T111342Z",
    "run_ts": "2026-04-29T11:13:42.777701+00:00",
    "phase": "initial",
    "project_root": "C:\\Users\\barath\\recomart-pipeline",
    "raw_root": "C:\\Users\\barath\\recomart-pipeline\\data\\raw",
    "bronze_root": "C:\\Users\\barath\\recomart-pipeline\\data\\bronze",
    "discovered_files": {
      "events_csv": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\retailrocket\\load_date=2026-04-29\\load_hour=11\\events.csv",
      "category_tree_csv": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\retailrocket\\load_date=2026-04-29\\load_hour=11\\category_tree.csv",
      "item_properties_part1_csv": "C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrock
et\\load_date=2026-04-29\\load_hour=11\\item_properties_part1.csv",
      "item_properties_part2_csv": "C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrock
et\\load_date=2026-04-29\\load_hour=11\\item_properties_part2.csv",
      "products_raw_json": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\dummyjson\\load_date=2026-04-29\\load_hour=11\\products_raw.json",
      "categories_raw_json": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\dummyjson\\load_date=2026-04-29\\load_hour=11\\categories_raw.json",
      "products_parquet": "C:\\Users\\barath\\recomart-
pipeline\\data\\bronze\\dummyjson\\products.parquet",
      "categories_parquet": "C:\\Users\\barath\\recomart-
pipeline\\data\\bronze\\dummyjson\\categories.parquet"
    },
    "overall_status": "FAIL",
    "issues_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_issues_initial_20260429T111342Z.csv",
    "summary_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_summary_initial_20260429T111342Z.csv",
    "datasets": [
      {
        "phase": "initial",
        "dataset": "retailrocket_events",
        "rows": 2756101,
        "columns": 5,
        "errors": 0,
        "warnings": 2734104,
        "status": "PASS"
      },
      {
        "phase": "initial",
        "dataset": "retailrocket_category_tree",
        "rows": 1669,
        "columns": 2,
        "errors": 0,
        "warnings": 25,
        "status": "PASS"
      },
      {
        "phase": "initial",
        "dataset": "retailrocket_item_properties",
        "rows": 20275902,
        "columns": 4,
        "errors": 0,
        "warnings": 0,
        "status": "PASS"
      },
      {
        "phase": "initial",

```

```

        "dataset": "dummyjson_products_raw",
        "rows": 194,
        "columns": 27,
        "errors": 0,
        "warnings": 92,
        "status": "PASS"
    },
    {
        "phase": "initial",
        "dataset": "dummyjson_categories_raw",
        "rows": 24,
        "columns": 3,
        "errors": 1,
        "warnings": 0,
        "status": "FAIL"
    },
    {
        "phase": "initial",
        "dataset": "products_parquet",
        "rows": 194,
        "columns": 32,
        "errors": 0,
        "warnings": 0,
        "status": "PASS"
    },
    {
        "phase": "initial",
        "dataset": "categories_parquet",
        "rows": 24,
        "columns": 8,
        "errors": 0,
        "warnings": 0,
        "status": "PASS"
    }
]
},
"revalidation": {
    "run_id": "20260429T111342Z",
    "run_ts": "2026-04-29T11:13:42.777701+00:00",
    "phase": "revalidat

```

14. Model Training Log Preview

```

{"event_ts": "2026-04-30T12:42:22.662312+00:00", "stage": "pipeline", "status": "started",
"message": "Fast model training notebook started", "extra": {}}
{"event_ts": "2026-04-30T12:42:57.919793+00:00", "stage": "pipeline", "status": "completed",
"message": "Fast model training notebook completed", "extra": {"rows_loaded_events": 196263,
"rows_loaded_products": 194, "rows_eval": 1500}}

```

15. Conclusion

This optimized notebook keeps the required deliverables while removing the main runtime bottlenecks from broad file scanning, oversized previews, full-volume evaluation, and unnecessarily heavy PDF rendering.